DHARSON R

QA Automation Engineer | SDET

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PROFESSIONAL SUMMARY

QA Automation Engineer transitioning from a manual testing background into SDET-focused roles, with hands-on experience building automated test generation platforms. Self-driven builder of multiple end-to-end QA automation tools combining Python, FastAPI, and Playwright. Comfortable across both backend automation and frontend UI development, with proven manual QA experience writing structured, production-style test case suites.

TECHNICAL SKILLS

- **Automation & Backend:** Python, FastAPI, Playwright, pytest
- **Frontend / Tools:** JavaScript, HTML/CSS, Streamlit, SheetJS, openpyxl (Excel export)
- **Testing & QA:** Manual functional testing, test case design, CRUD/persistence testing, cross-platform (iOS/Android) test comparison, Advanced Excel
- **Data & Storage:** ChromaDB, Firebase Firestore
- **Deployment:** GitHub, Streamlit Cloud

QA / SDET PROJECTS

TestCraft — Automated QA Testing Platform

- Built a platform that scans live websites, automatically generates test cases, and runs CRUD persistence checks against the live site using Playwright.
- Architected the system across a FastAPI backend, a Playwright-based CRUD testing engine, and a JavaScript/HTML frontend.
- Currently driving fixes to authenticated-flow scanning (role-picker login handling) and session-stability issues in site-wide test cycles.
- **Earlier prototype — QAVision:** Built a lightweight single-file (HTML/CSS/JS) precursor combining live URL scanning, automated test case generation, and Excel/Playwright export across 8 iterations — the groundwork that informed TestCraft's design.

Automated Test Generation & Self-Healing Agent

- Built an end-to-end automated testing agent from scratch using Python, pytest, ChromaDB for vector-based memory storage, and Streamlit.
- Implemented automatic test generation, bug detection, and automatic code fixing, with a live Agent Summary UI.
- Extended the agent for multi-language test case generation (Python through PHP) with one-click Excel export via openpyxl; deployed publicly on GitHub and Streamlit Cloud.

MANUAL QA TESTING EXPERIENCE

- **Matrimony Web & Mobile Application (Independent QA Practice):** Authored 110+ real-world test cases across 22 modules for a matrimony web application, and a 75-case iOS-vs-Android comparison suite across 6 scenarios in a formatted, color-coded Excel workbook (STATUS/PRIORITY tracking).

WORK EXPERIENCE

Telecaller | *Nithra Apps Private Limited* Dec 2024 – Present

EDUCATION

- **B.E., Electronics and Communication Engineering —** K S Rangasamy College of Technology, Anna University (77.5%)
- **Diploma in Electronics and Communication (DECE) —** KSR Polytechnic College, DOTE University (80%)
- **HSC —** Government Boys Higher Secondary School, Bavani (51%)
- **SSLC —** Government Boys Higher Secondary School, Komarapalayam (71%)

FREELANCE WEB DEVELOPMENT (CLIENT PROJECTS)

- **Event Stall Tracker:** Built a Firebase Firestore-backed event management web app with employee PIN login, real-time token/entry tracking, paid/unpaid status, CSV & image bulk import, an admin leaderboard, and a soft-delete recycle bin.
- **Kaviya Fancy Store:** Built a React-based POS and inventory management dashboard for a retail store client, with stock tracking and low-stock alerts, a sales cart with profit calculation, daily reports, and CSV import/export.

OTHER PROJECTS

- **Assistive Device for Blind Individuals Using Deep Learning:** Raspberry Pi + OpenCV/TensorFlow facial recognition device with real-time voice alerts (95% accuracy); patented (Patent No. 202441036456, issued 01.06.2024).

CERTIFICATIONS & COURSES

- NPTEL — IoT: Internet of Things (September 2022)
- ICT Academy Learnathon 2021 — Cybersecurity Essentials, Image Processing, RPA Essentials, Business Analyst, Network Visualization

LANGUAGES

Tamil, English